Shock Breakout

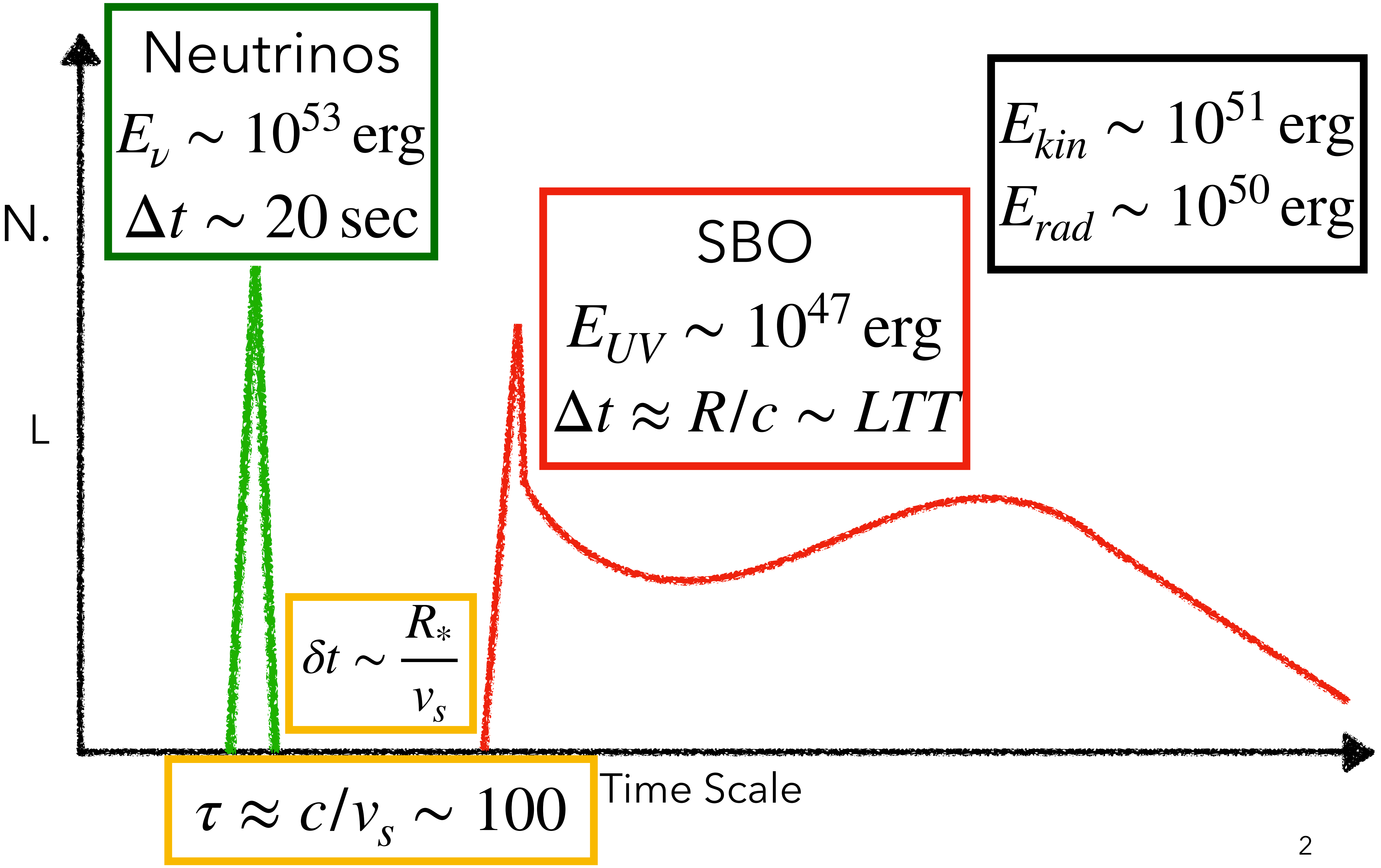
SN1987A/ RSG

17.09.25 wychen

Shock Breakout

Reasoning

- First Electromagnetic Signals of SN.
- Shocks propagate from optically thick to optically thin.
- Need Rad-Hydro+Shock



- D. Nadyozhin and V. S.imshennik. Physics of supernovae. *International Journal of Modern Physics A*, 20, 01 2012.
- Waxman, E., & Katz, B. 2017, in Handbook of Supernovae, ed. A. W. Alsabti & P. Murdin (Springer, Cham)

Progenitor

Explosion

Shock Breakout

CSM Interaction

Observer

Progenitor

Shock Breakout

Observer

BSG SN1987a

2D

UV Tail

SN1987a Shock Breakout

Chen W.-Y., Ke-Jung Chen, Masaomi Ono, 2024, ApJ, 976, 147.

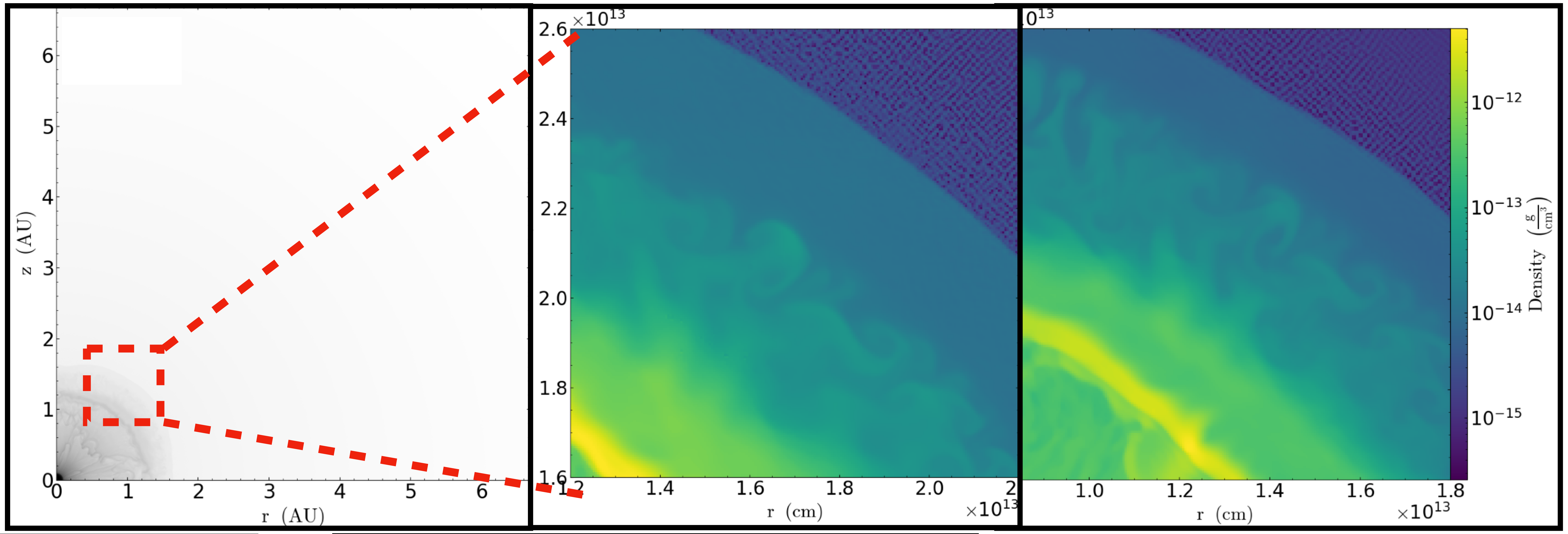
CASTRO

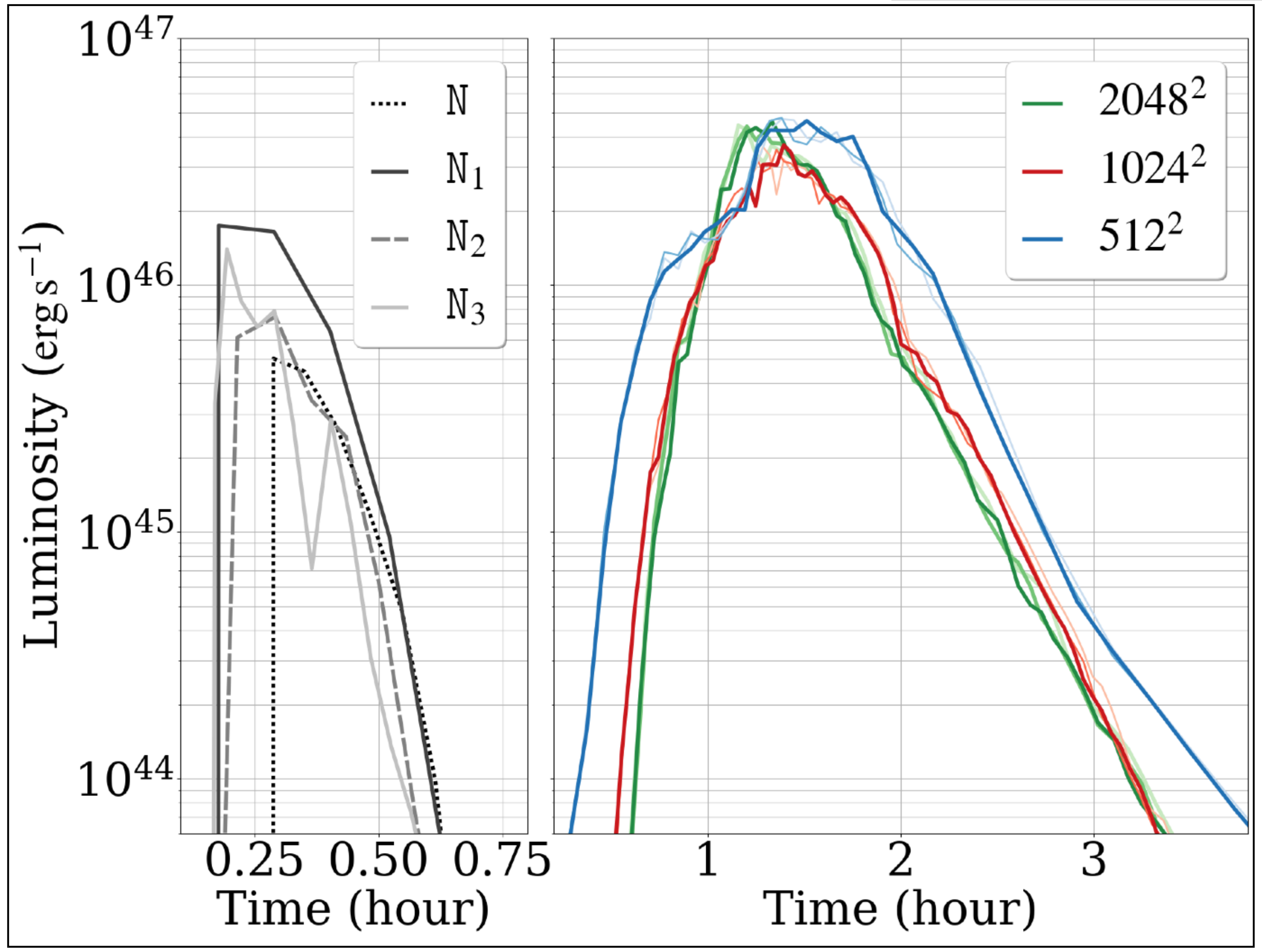
Almgren, Beckner, Zhang, Howell, Katz, Zingale. 2010-2020

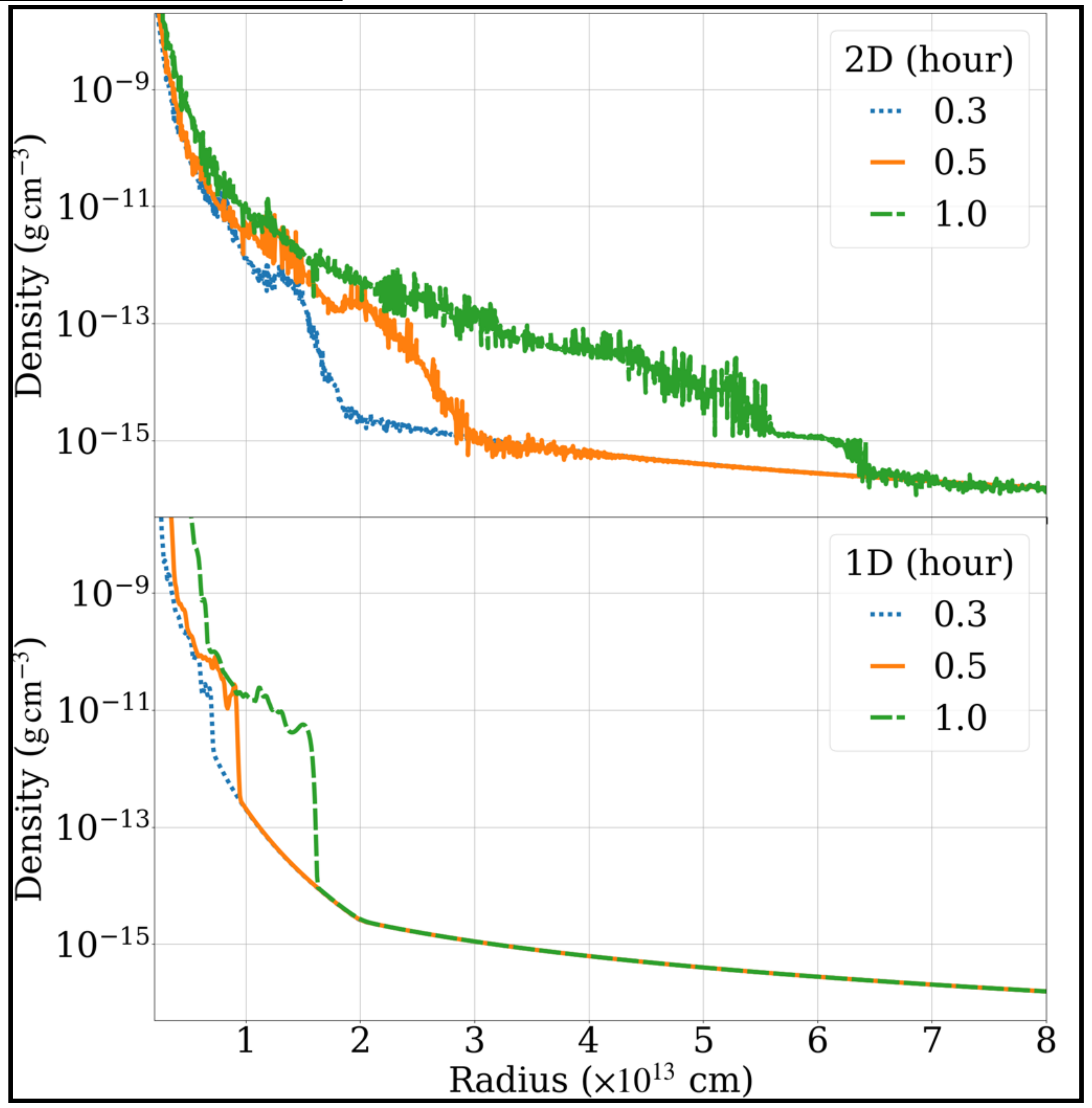
- AMR-based, compressible hydrodynamics code. Support MHD, general equations of state, full Poisson equation, and reactive networks.
- MPI + OpenMP for CPU architectures, MPI + CUDA for NVIDIA GPUs, and MPI + HIP for AMD GPUs
- MGFLD:
  - In optically thick regimes, MGFLD reduces to classical diffusion.
  - In optically thin regions, radiation behaves as free-streaming.
  - In the intermediate regime flux limiters.
  - Frequency-dependent.

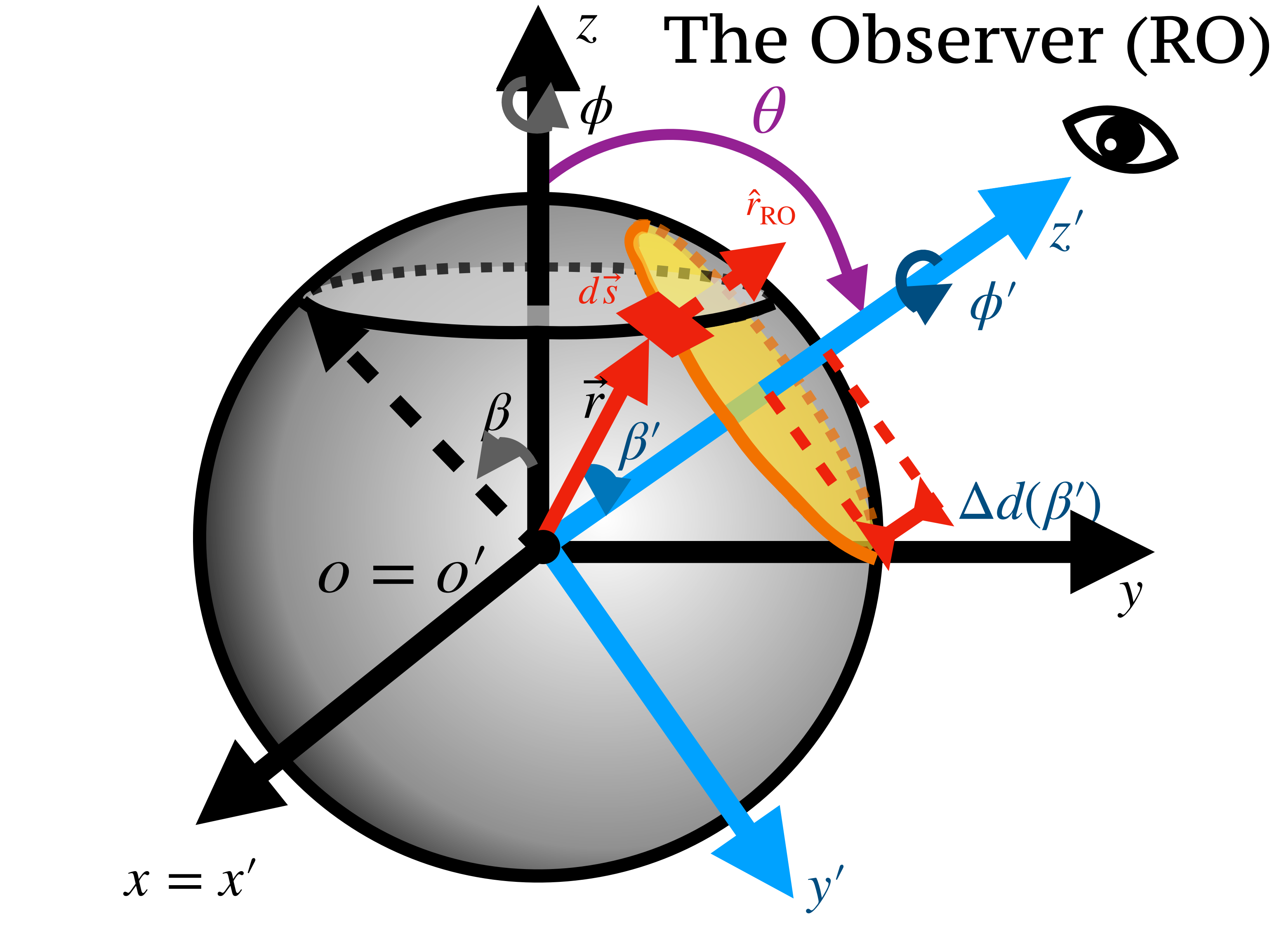
2D 1987a

- Chen W.-Y., Ke-Jung Chen, Masaomi Ono, 2024, ApJ, 976, 147.
- Structures and LC
- Spherical Progenitor





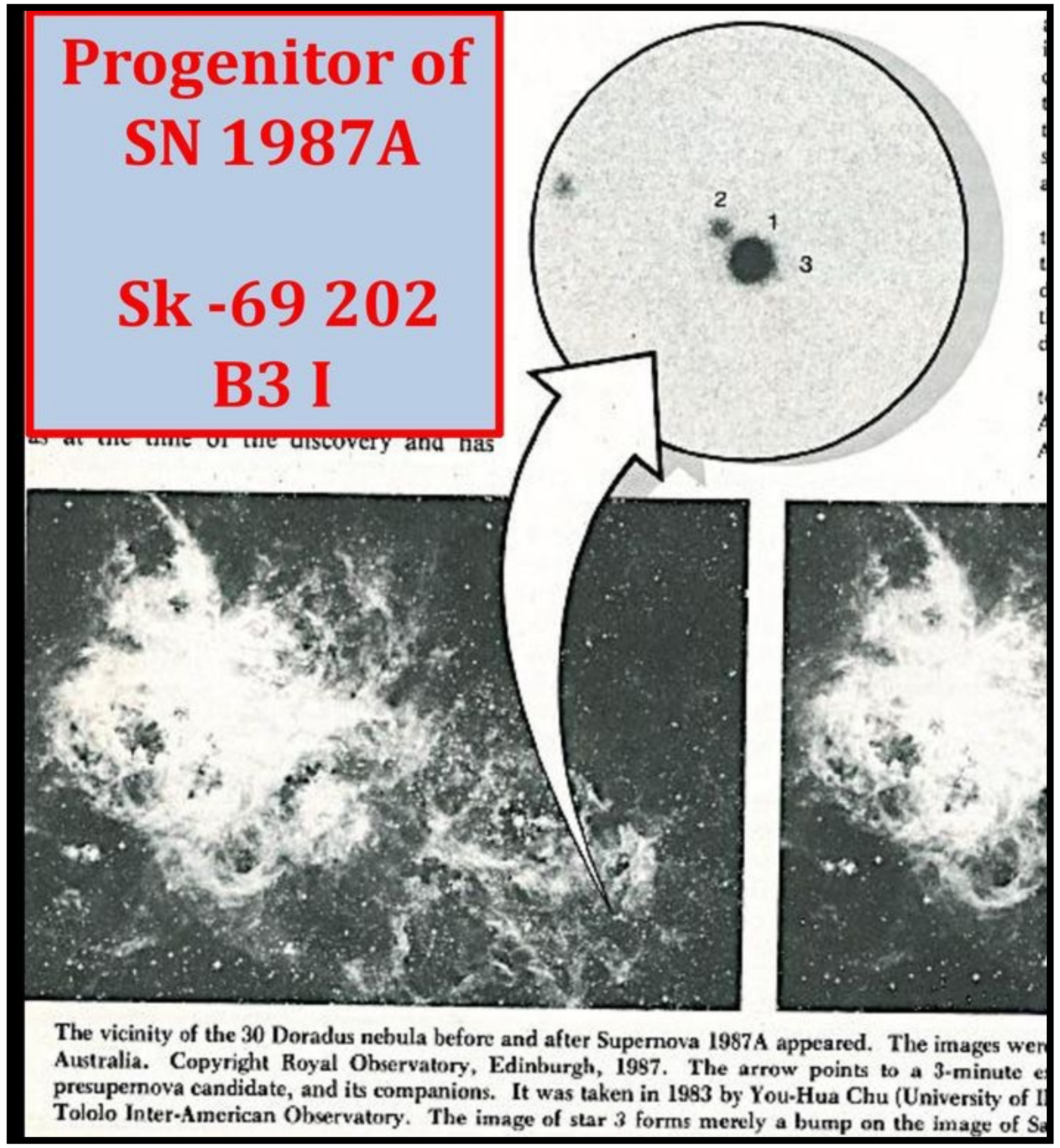




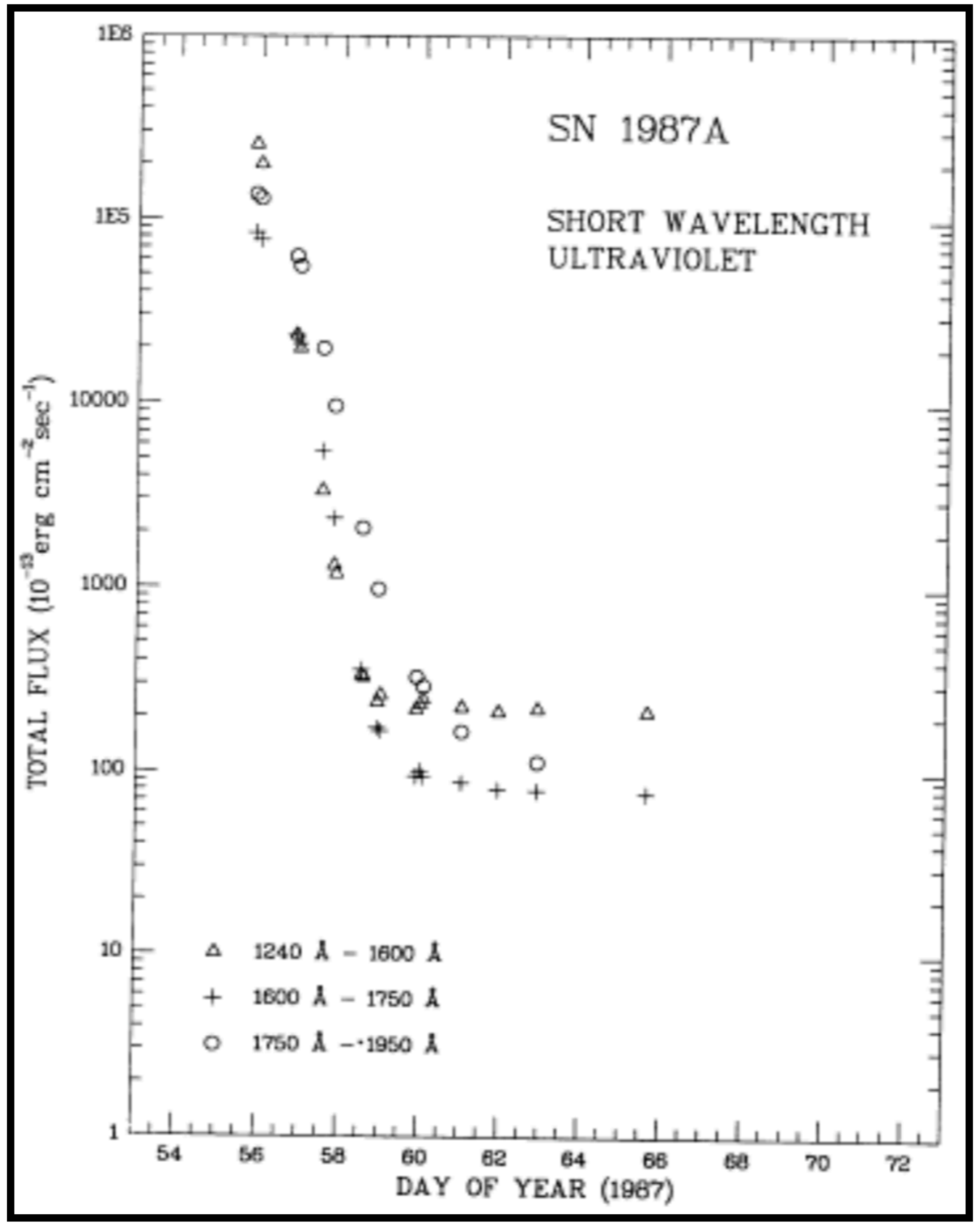
At $|\vec{r}| = r_s = 8 \times 10^{13}$ cm

Early Observation

- You-Hua Chu: SK -69 202 B31
- Sonneborn and Kirshner (1987);
  Wamsteker (1987)
- The earliest most observation of UV is done after 13 hours of the discovery.
- Only the “tail” of shock breakout in UV can be compared: that drop from 10^7 K to 10^4 within 1-2 days.



The vicinity of the 30 Doradus nebula before and after Supernova 1987A appeared. The images wer
Australia. Copyright Royal Observatory, Edinburgh, 1987. The arrow points to a 3-minute e
presupernova candidate, and its companions. It was taken in 1983 by You-Hua Chu (University of I
Tololo Inter-American Observatory. The image of star 3 forms merely a bump on the image of Sa



Progenitor

Shock Breakout

Observer

BSG SN1987a

2D ~hour

UV Tail

Progenitor

Explosion

Shock Breakout

CSM Interaction

Observer

Progenitor	Shock Breakout	Observer
RSG 20 & 25	2D hour-day	Stronger Mass Loss

RSG Shock Breakout

In Prep…

Shock Breakout

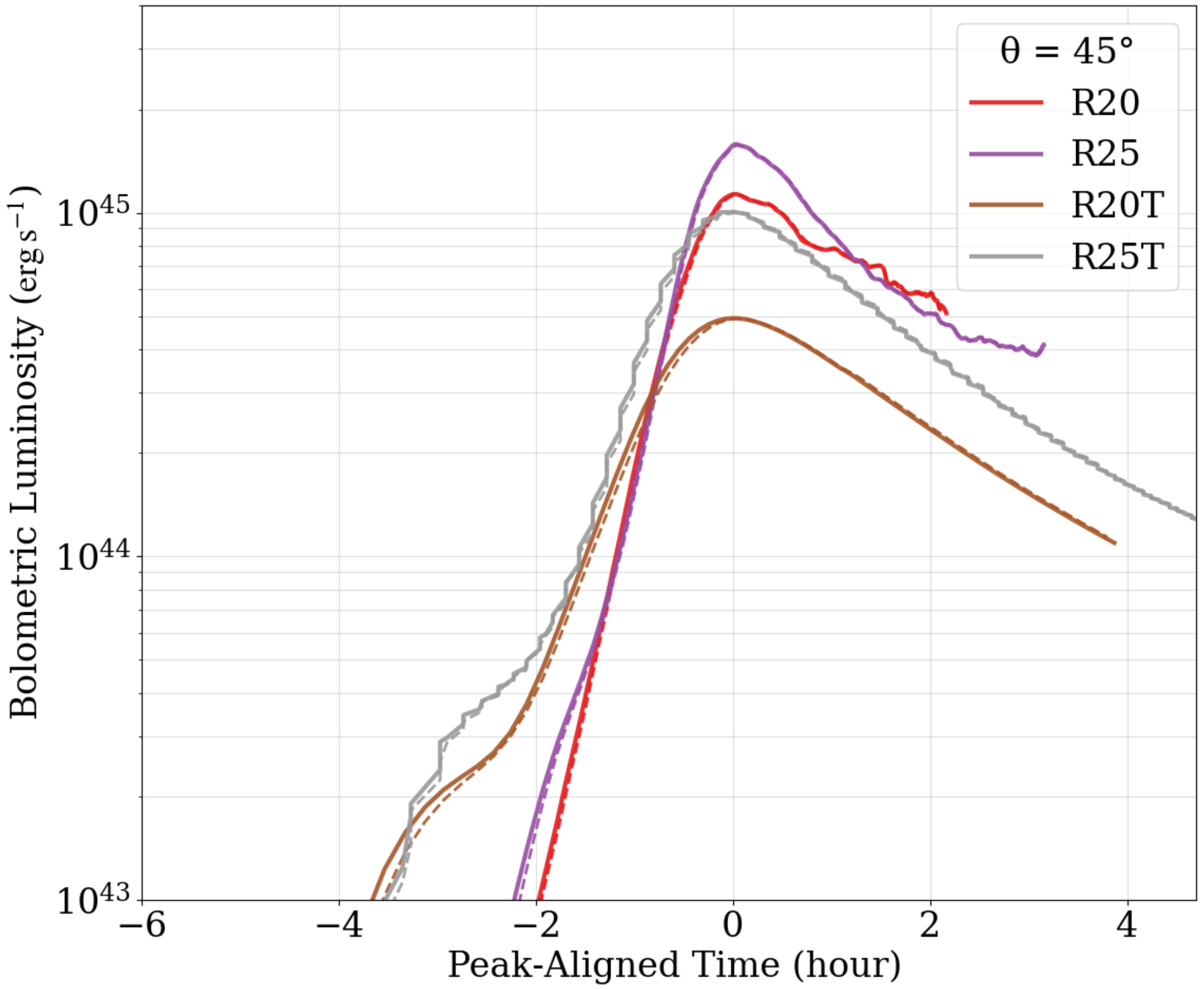
RSG

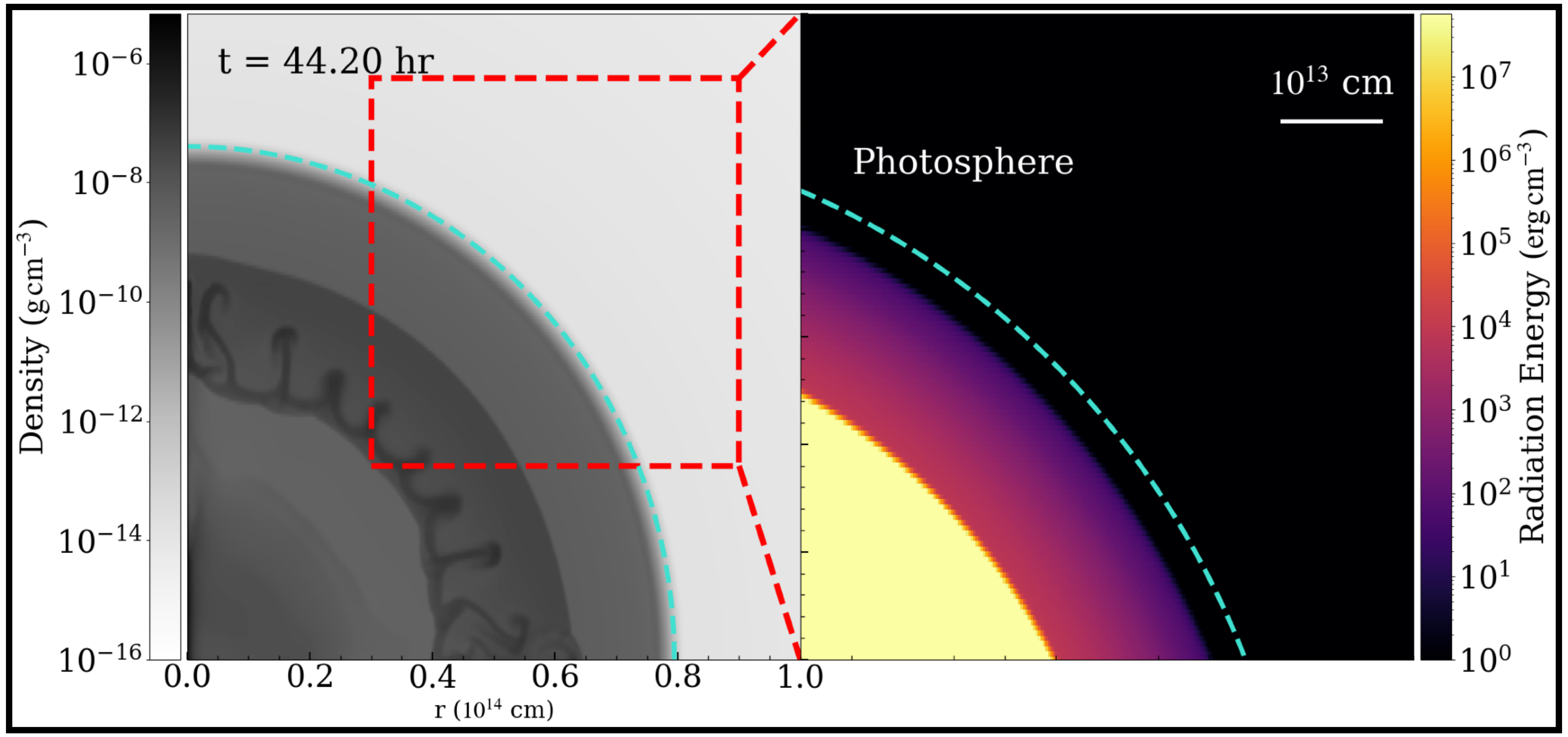
- Longer than BSG and Thicker CSM profile
- Aim at distinguishing the effects from multi-D, radiation precursors, progenitors, and the CSM profile.
- Shock Breakout structures revisited

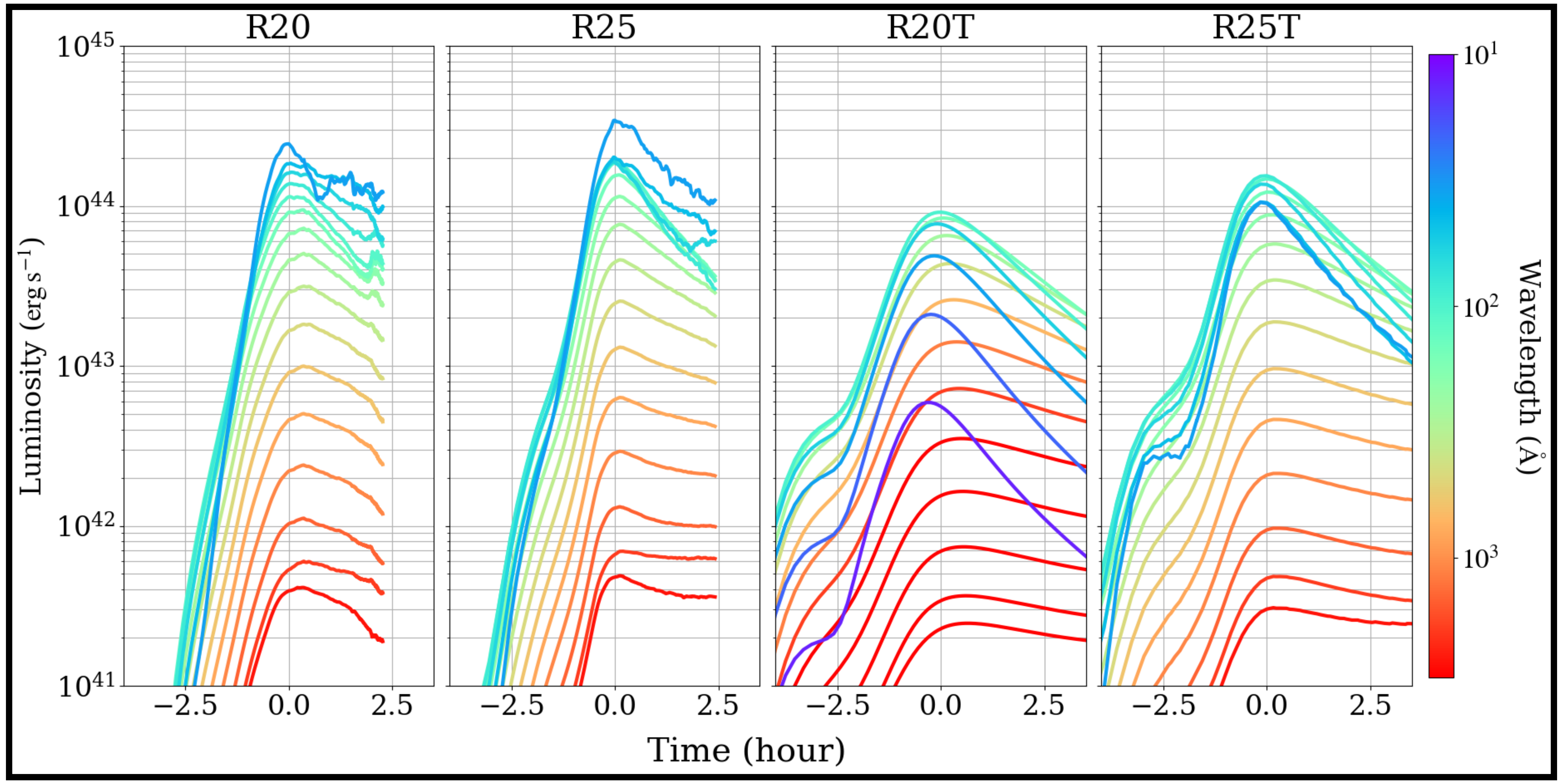
Shock Breakout

Later Time

- Double Peak
- Shock Breakout (RPS) always Interacting SNe?



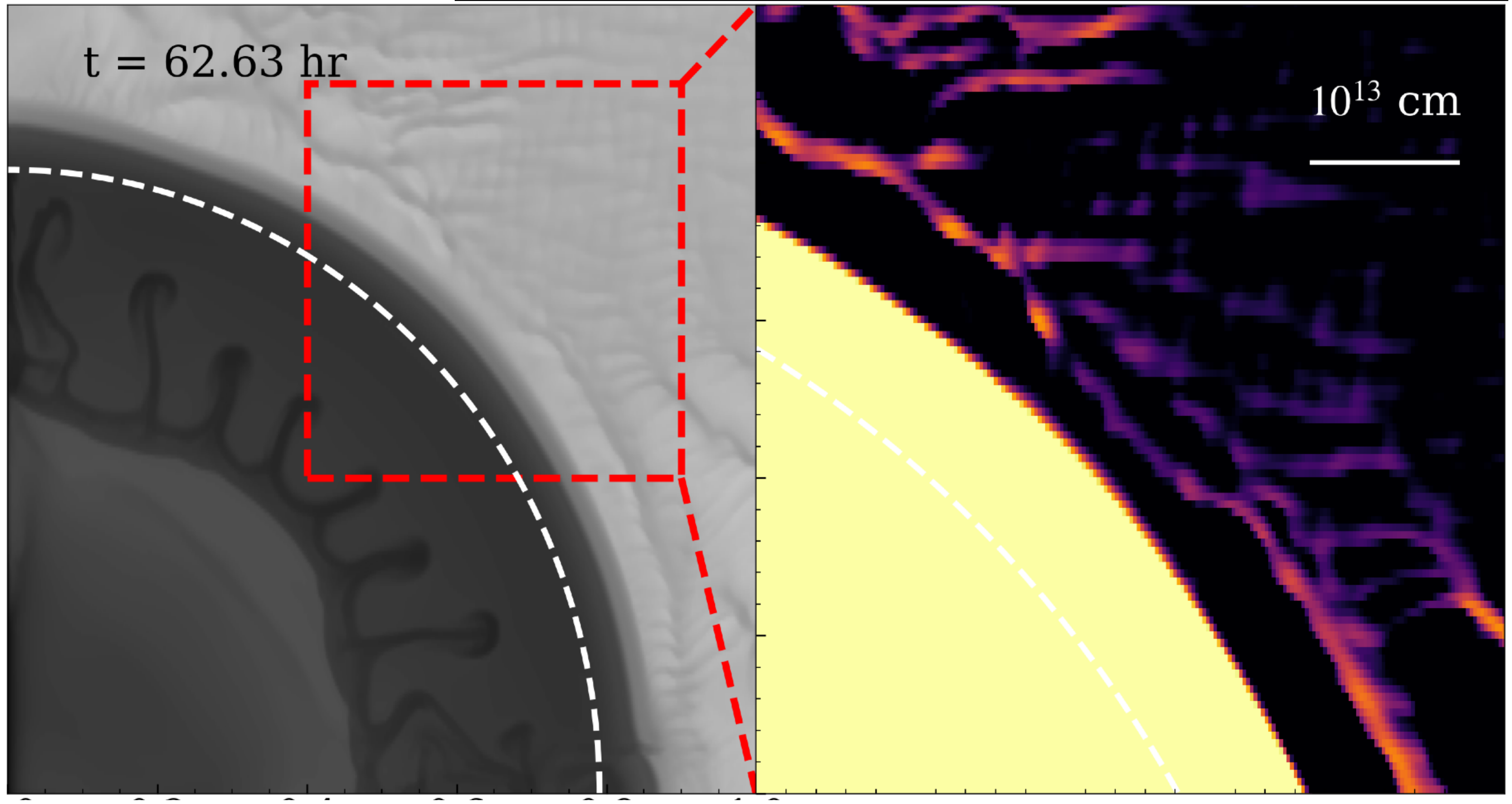


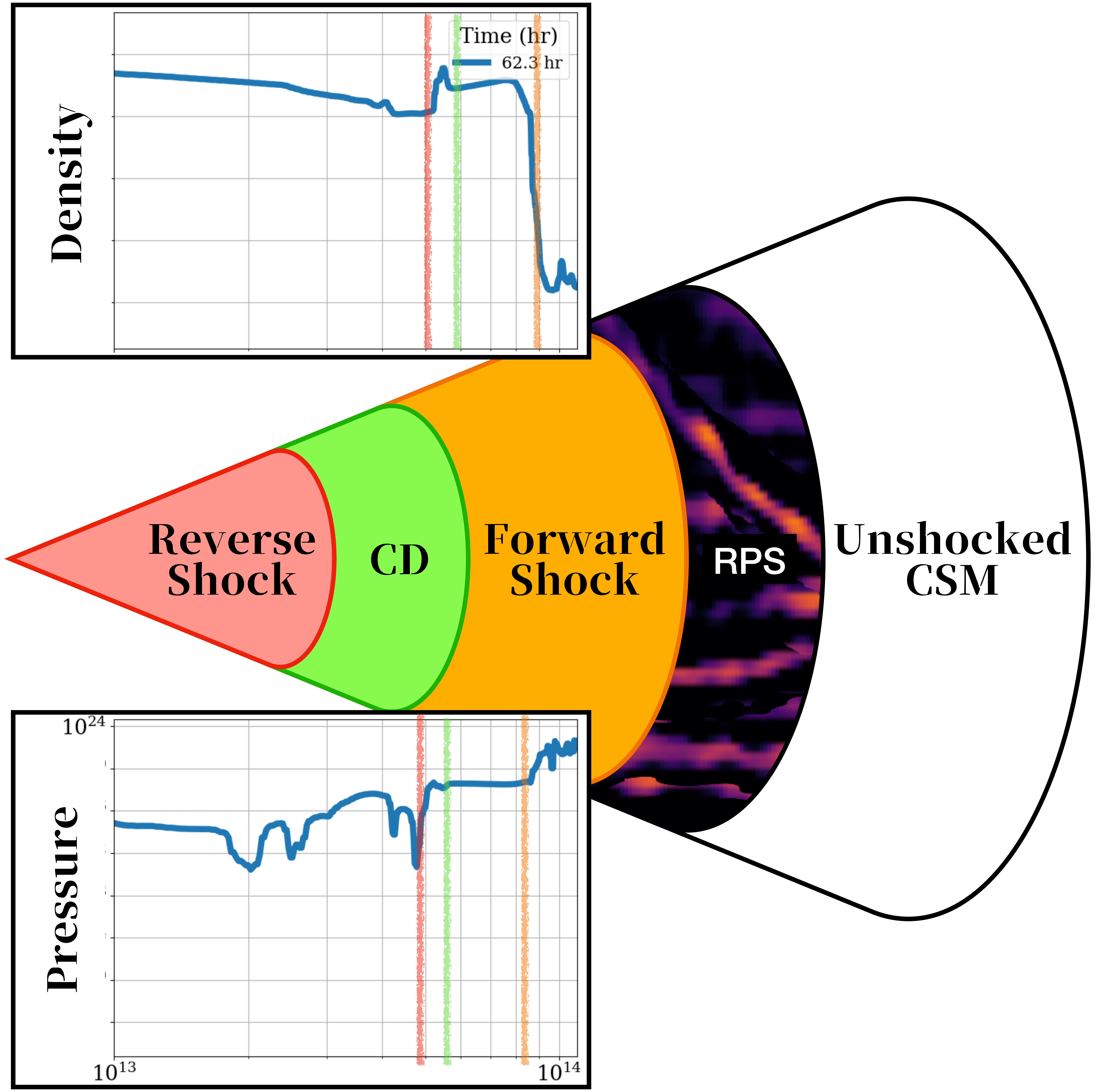


Rad-Precursor

CSM Interaction

- RS/ FS: Pressure Rise.
- CD: Density gradient without pressure rise/ drop.
- RPS: Infront of FS (C-shock)





Atmosphere

XRT 080109/ SN 2008D

- Takashi, Debashis, **Norbert.** 2015
  - Eddington Limit and inflated envelopes.
  - Diffusion > Light Crossing Time

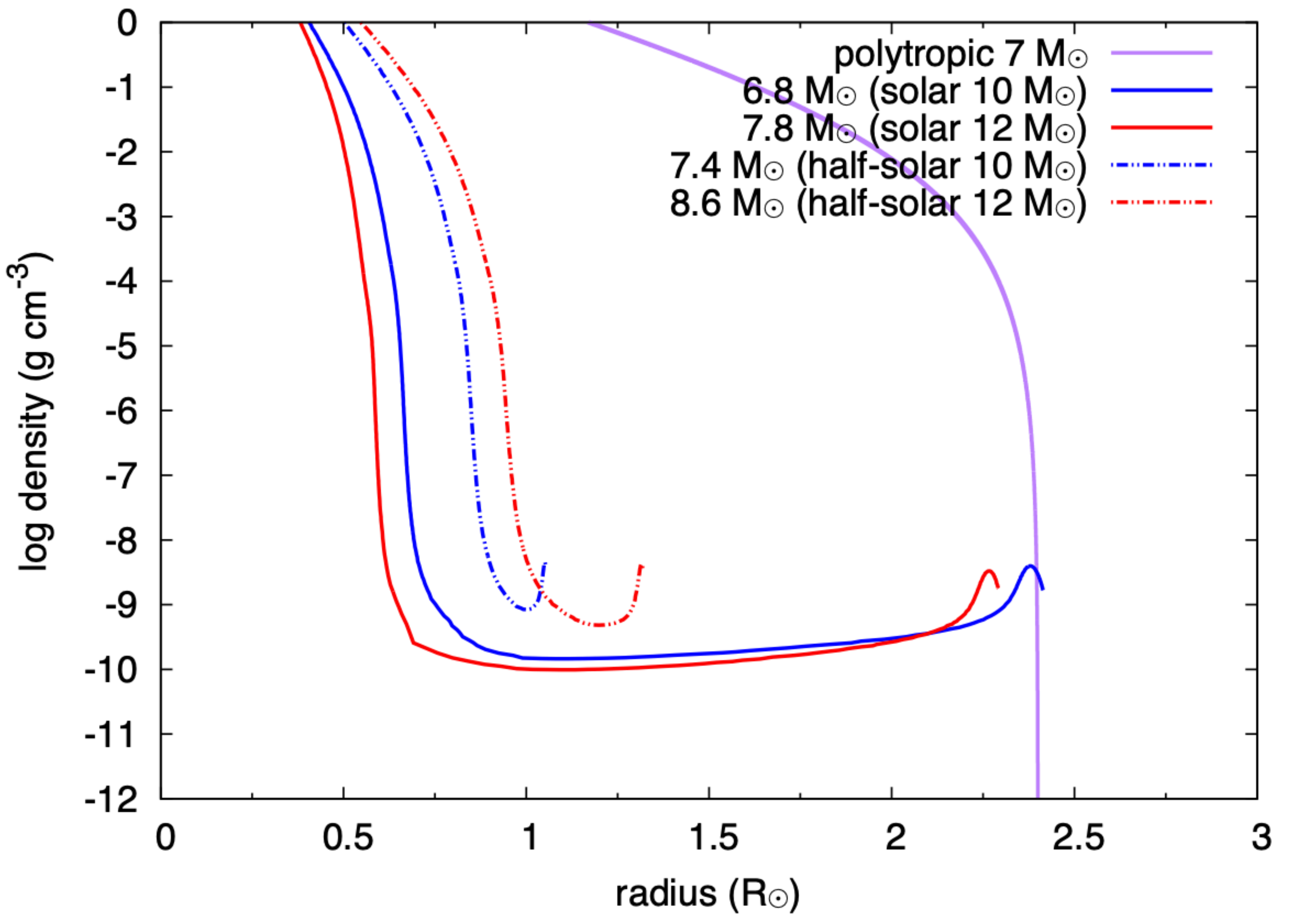
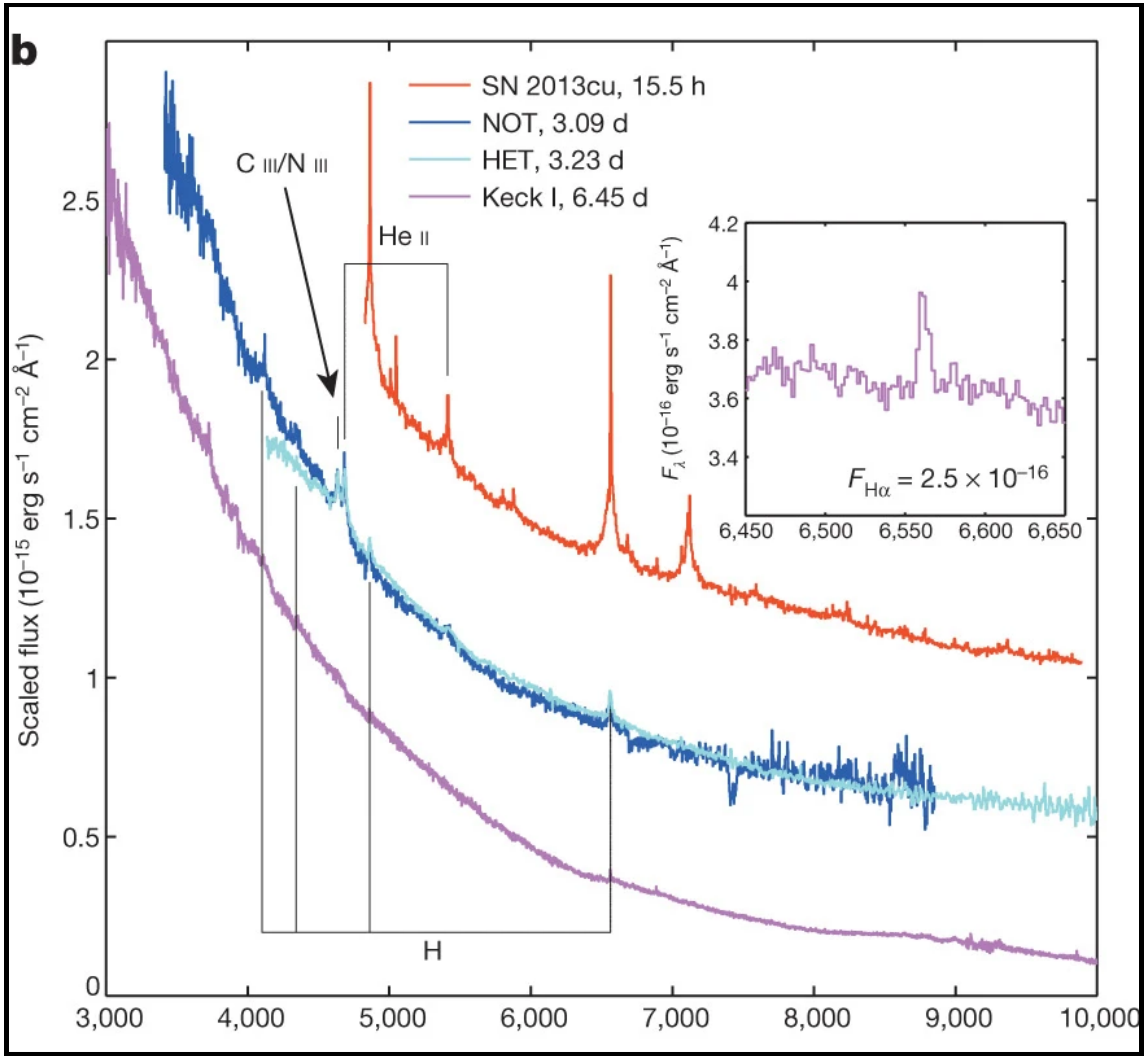


Fig. 2. Final density structure of our stellar models. We also show the structure of a star without the envelope inflation, which is a polytiropic star of 7 $M_{\odot}$ and 2.4 $R_{\odot}$ with the polytropic index of 3. The inflated stellar models have the extended low-density regions on top of the core structure.

Stellar Wind

SN 2013cu

- Gal-Yam, A., Arcavi, I., Ofek, E…, 2014
  - WR signals becomes featureless after ~ 6 days of explosion.
- Graefener and Vink 2016
  - Not a WR wind if include LTT effects
  - After 15.5 hours of explosion



Shock Breakout

Take Home Messages

- Shock Breakout ~ Instead of the light traveling time, it is a continuous process that starts at radiation precursor and ends by releasing radiation.
- Multi-D effects + RHD ~ Radiation Precursor that signals the beginning of shock breakout.
- MGFLD ~ Separate the progenitors and CSM impacts on the LCs.

Progenitor

RSG

20 & 25

Shock

Breakout

2D

hour-day

Observer

Stronger

Mass Loss

Progenitor

Explosion

Shock Breakout

CSM Interaction

Observer

Progenitor	Shock Breakout	CSM Interaction	Observer
RSG 20 & 25	2D hour-day	Confined-Shell CSM	Yaron 2017

Confined-Shell CSM

HITS Summer Institute Programme

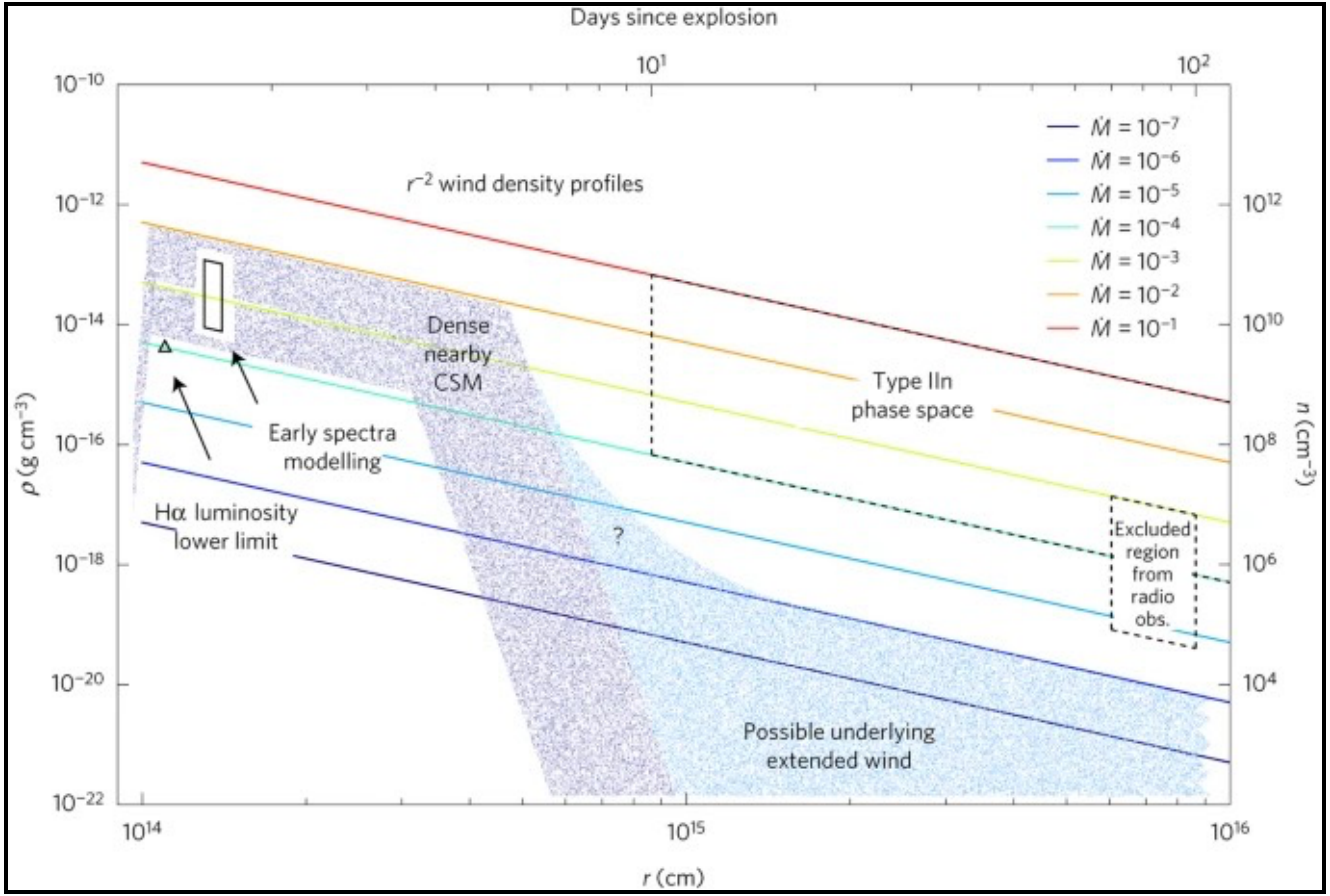
Conference in EAMA 11, Japan

Shock Breakout

Confined-Shell CSM

- Can constraint Mass Loss Rate?
- Need HD simulations on CSM formation
- Common Envelope?

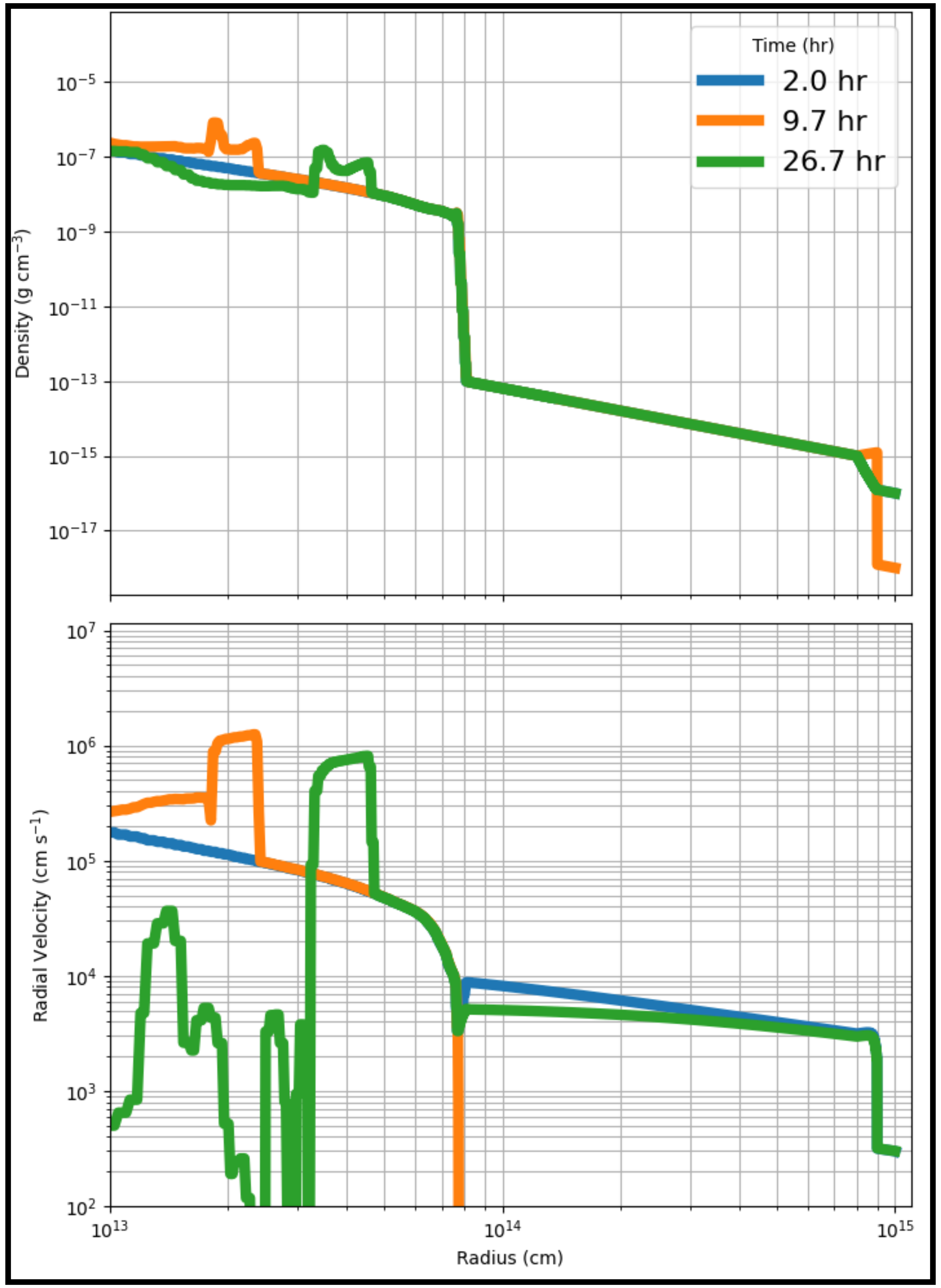
Rad-Precursor



Yaron, O., Perley, D., Gal-Yam, A. *et al.* Confined dense circumstellar material surrounding a regular type II supernova. *Nature Phys* **13**, 510–517 (2017). https://doi.org/10.1038/nphys4025

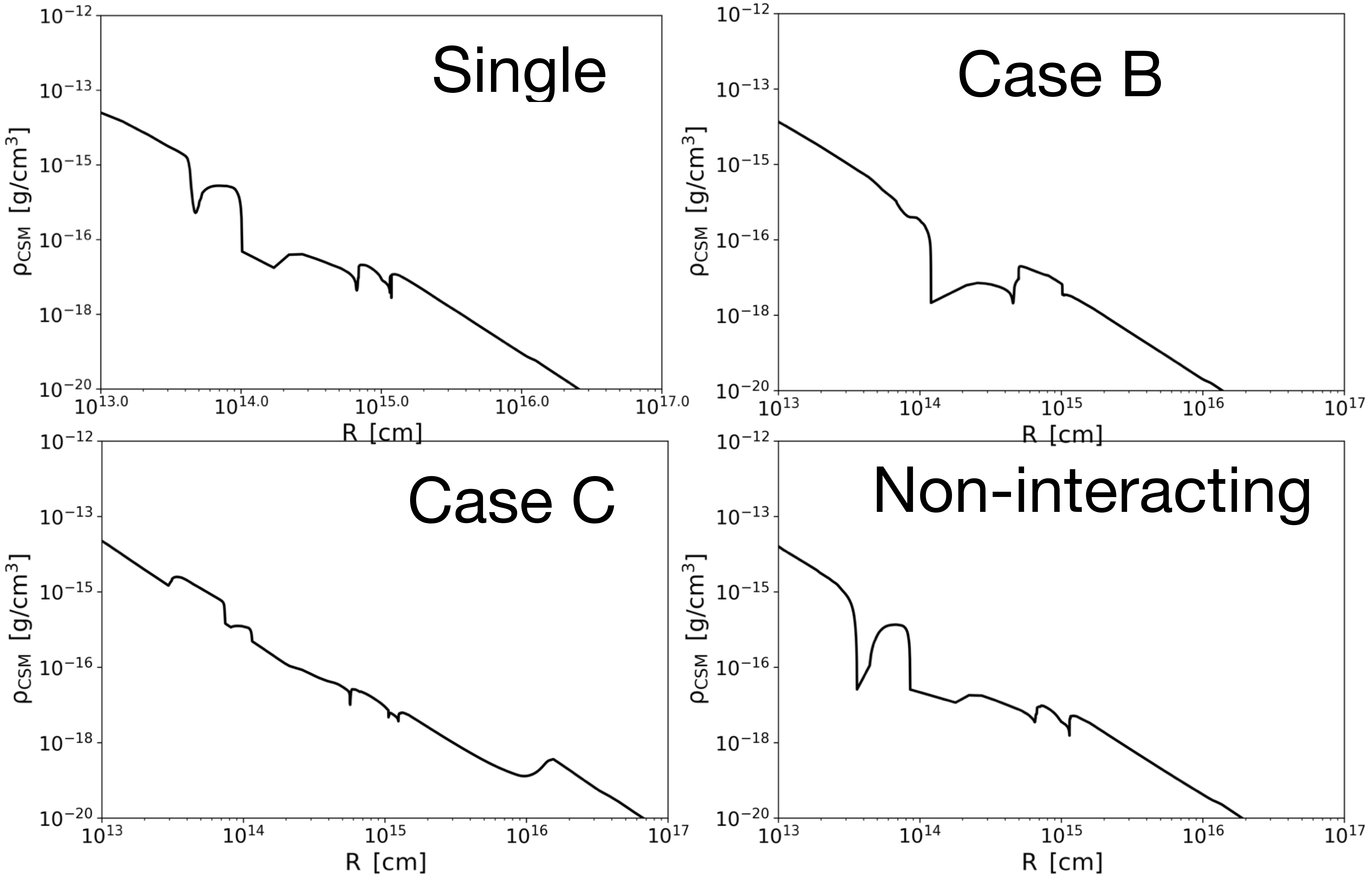
1. Observation

- Setup from Yaron 2017
- Use density ~ $\rho \approx \dfrac{\dot{M}}{4\pi r^2 v}$
- Have degeneracy of mass loss rate and wind velocity
- Time evolution is also degenerate.
- Testing other smoothing.



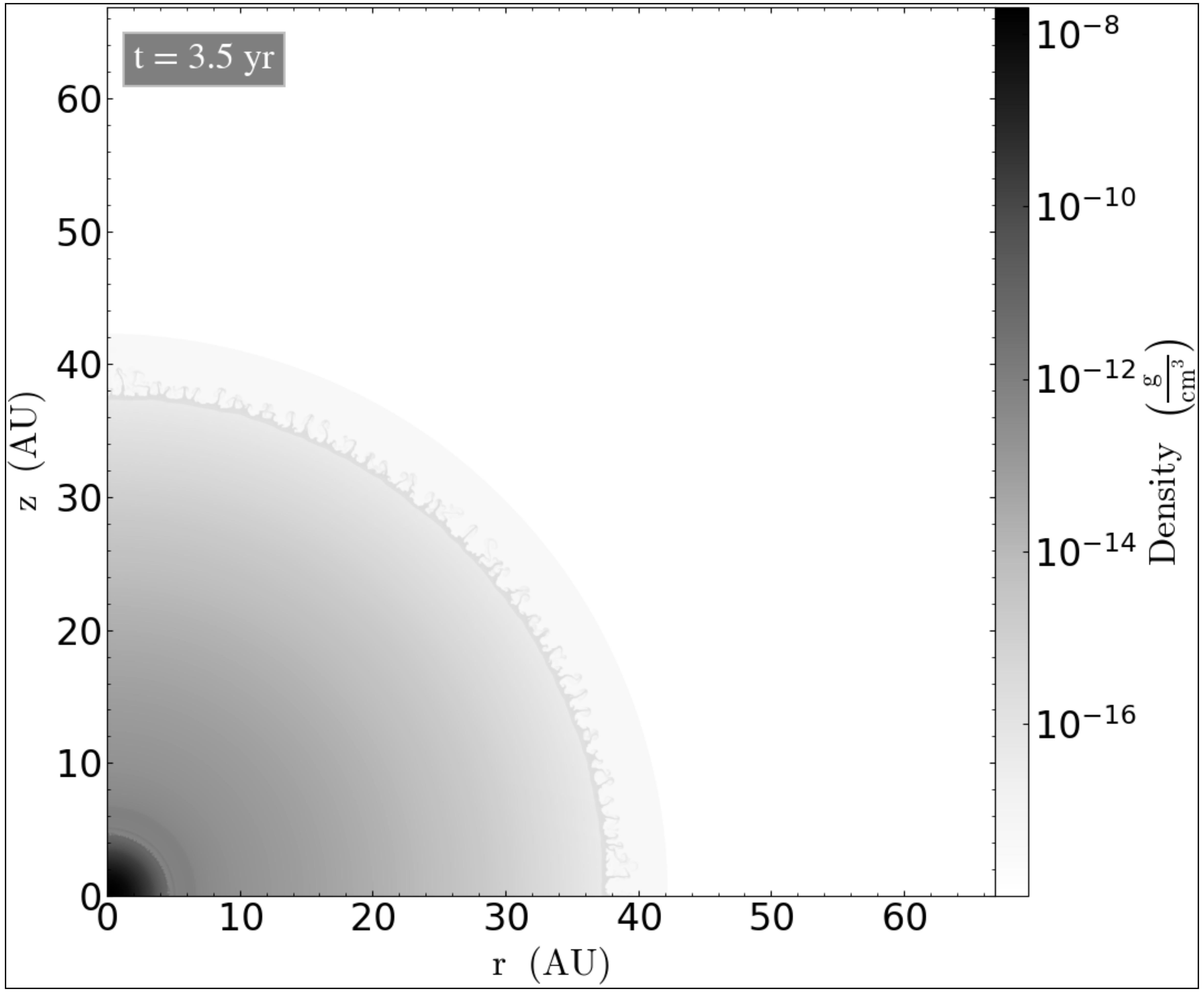
2. Sung-Han's

- Setup from Sung-Han's Model
- Use density ~ $\rho \approx \dfrac{\dot{M}}{4\pi r^2 v}$
- Weird Cut, Require Binning
- Depends on Explposion Timing



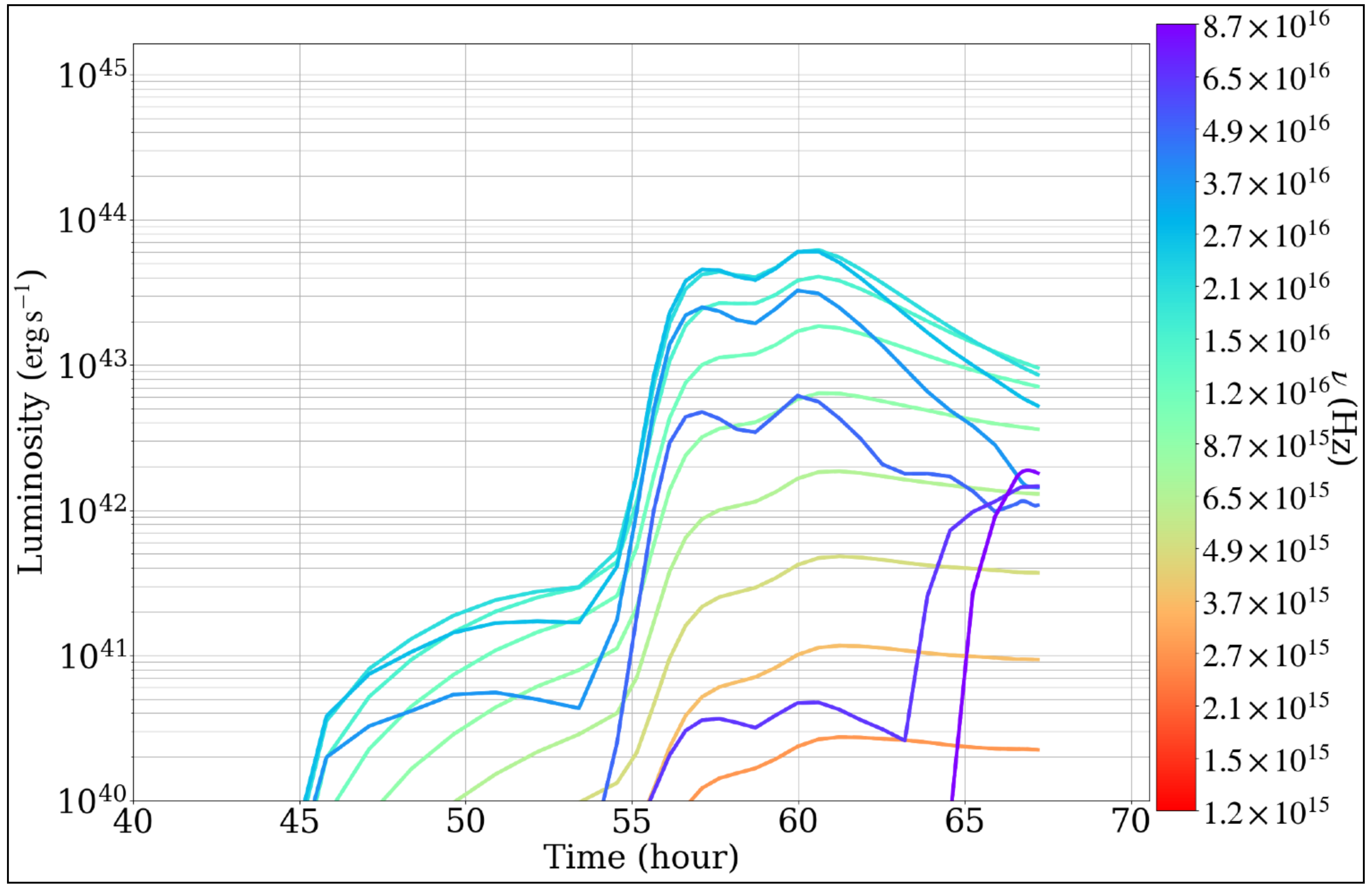
3. CSM HD

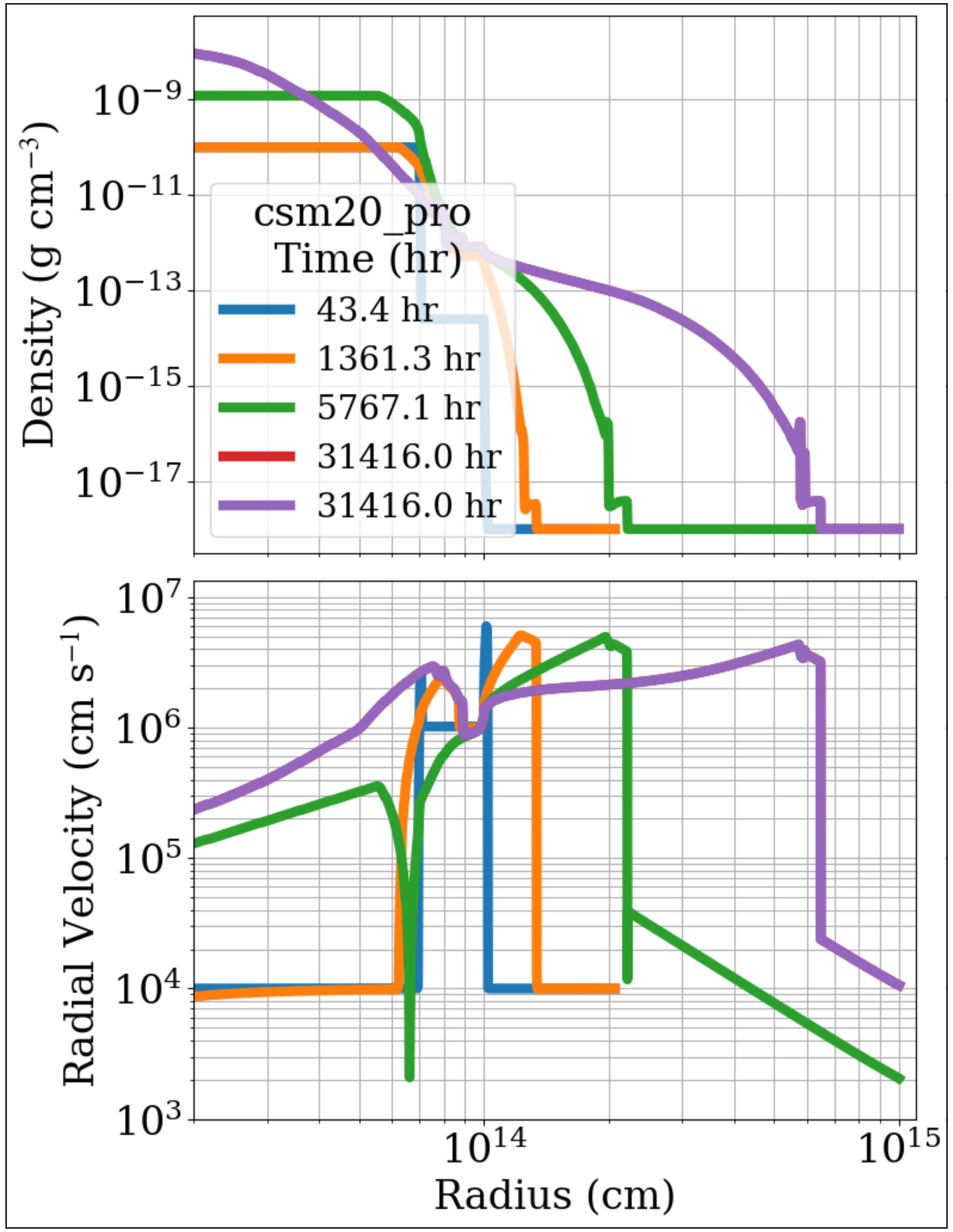
- HITS Group (Mike, Vetter …)
- Tomoki: Long Term HD CSM:
  - https://arxiv.org/pdf/2504.14255
- Input mass flux from stellar radii
- Crop 3yr/ 10yr/ 100yr before SN till the wind blows to > 60 AU
- All outflow are spherical.



3. CSM HD

- Tool:
    - Common Envelope Mass Loss
    - Jing Ze (Ejected/ Saturated) MPA
    - 2013cu (Observation) Bonn
    - Extended Stellar Envelope Bonn





Progenitor	Shock Breakout	CSM Interaction	Observer
RSG 20 & 25	2D hour-day	Confined-Shell CSM	Yaron 2017

Progenitor

Explosion

Shock Breakout

CSM Interaction

Observer